

UCT202: STATISTICAL MODELING

L	T	P	Cr
3	0	2	4.0

Course Objectives: To provide an understanding of various statistical methods such as correlation analysis, regression, sampling, estimation, non-parametric inference, time-series analysis and forecasting.

Sampling Techniques: Random sampling, Sampling from finite and infinite populations, Estimates and standard error (sampling with replacement and sampling without replacement), Sampling distribution of sample mean, stratified random sampling.

Linear Statistical Models: Scatter diagram. Linear regression and correlation, least squares method, Rank correlation, Multiple regression & multiple correlation, Analysis of variance (one way, two way with as well as without interaction).

Estimation: Point estimation, criteria for good estimates (un-biasedness, consistency), Methods of estimation including maximum likelihood estimation.

Sufficient Statistic: Concept & examples, complete sufficiency, their application in estimation.

Test of hypothesis: Concept & formulation, Type I and Type II errors, Neyman Pearson lemma, Procedures of testing.

Non-parametric Inference: Comparison with parametric inference, Use of order statistics, Sign test, Wilcoxon signed rank test, Mann-Whitney test, Run test, Kolmogorov-Smirnov test, Spearman's and Kendall's test, Tolerance region.

Basics of Time Series Analysis & Forecasting: Stationary, ARIMA Models: Identification, Estimation and Forecasting.

Laboratory Work:

R statistical programming language: Introduction to R, Functions, Control flow and Loops, working with Vectors and Matrices, reading in Data, writing Data, working with Data, Manipulating Data, Simulation, Linear model, Data Frame, Graphics in R

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Comprehend different sampling techniques for designing and selecting a sample from a population.
2. Apply and interpret basic modeling techniques for bivariate data and use inferential methods in the context of simple linear models.
3. Understand the estimation of finite population parameters e.g. mean and variance including maximum likelihood estimation.

4. Comprehend the basics of hypothesis testing and perform different non-parametric tests.
5. Illustrate and apply the important features to describe a time series and perform simple analysis and computations on series.

Text Books:

1. Miller I.R., Freund J.E. and Johnson R., Probability and Statistics for Engineers, Pearson Education India (2015) 8th ed.
2. Goon A.M., Gupta M.K. and Dasgupta B., Fundamentals of Statistics, Vol. I & II, The World Press (2002), 8th ed.
3. Chatfield C., The Analysis of Time Series: An Introduction, Chapman & Hall/CRC (2003) 6th ed.

Reference Books:

1. Montgomery D.C. and Peck E., Introduction to Linear Regression Analysis, Wiley-Interscience (2006), 4th ed.
2. Mood A.M., Graybill F.A. and Boes D.C. Introduction to the Theory of Statistics, McGraw Hill (2017), 4th ed.
3. Draper N. and Smith H., Applied Regression Analysis, Wiley-Interscience (1998), 3rd ed.
4. Golemund G., Hands-on Programming with R, O'Reilly (2014).
5. Lander J.P., R for Everyone: Advanced Analytics and Graphics, Addison-Wesley Professional (2017), 2nd ed.